

	Case Name: Industrial Complex (1)	Sector	Construction (industrial)	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
Submitted by	Matilde Casado and Margarida Antunes		One of nine std. scenarios	
Date			User defined distributions	
File Name	C2015-03 Industrial Complex (1)	Project Control	Automatic tracking	
			Tracking based on user input	

1. Project description

Project authenticity

The project includes foundation work, structural assembly, masonry, roofing, drainage, facade installations, and interior finishes such as HVAC, electrical, sanitation, flooring, and painting, ensuring a fully functional and modern facility.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	135	Serial/Parallel (SP)	16%
Planned Duration (PD)	257 days*	Activity Distribution (AD)	43%
Budget At Completion (BAC)	2 244 090.7€	Length of Arcs (LA)	0%
Renewable Resources	-	Topological Float (TF)	58%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity				Time sensitivity		
	avg [%]	std dev [%]	skew [-]		avg [%]	std dev [%]	skew [-]
CRI-r	9.1	9.4	-2.9	CI	10.3	13.2	-1.2
CRI-rho	11.9	13.2	-2.2	SI	27.4	22.0	-0.6
CRI-tau	12.4	22.8	-3.4	SSI	3.3	7.1	-3.5
				CRI-r	9.1	8.4	-2.1
				CRI-rho	12.2	12.5	-2.0
				CRI-tau	13.0	22.7	-3.4
	Resource sensitivity						
	avg [%]	std dev [%]	skew [-]				
CRI-r	N/A	N/A	N/A				
CRI-rho	N/A	N/A	N/A				
CRI-tau	N/A	N/A	N/A				

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	24.6	-13.5	1	1.6	0.2
PV - SPI	50.1	50.1	CPI	3.6	0.5
PV - SCI	50.1	50.0	SPI	28.3	28.3
ED - 1	115.7	114.6	SPI(t)	23.9	23.9
ED - SPI	50.1	50.1	SCI	28.3	28.3
ED - SCI	50.1	50.0	SCI(t)	23.9	23.9
ES - 1	25.9	-23.1	0.8 CPI + 0.2 SPI	20.8	20.8
ES - SPI(t)	27.0	26.3	0.8 CPI + 0.2 SPI(t)	12.5	12.5
ES - SCI(t)	27.0	26.3			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 9 tracking periods with a length of approximately 1 month week(s). The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	249030	-103590.6	-83.9	1.3	0.9	0.9	1.0
std dev	106860.2	142419.6	88.5	0.2	0.3	0.1	0.0
final	369050	-6245.3	-202	1.2	1.0	0.9	1.0

2.3.3.2. Time forecasting

PD	257 days	Real Duration	282 days	Late	9.7%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	260.0	1.9	0.7	0.7
PV - SPI	260.8	1.9	0.7	0.7
PV - SCI	261.6	2.1	0.7	0.7
ED - 1	262.2	2.1	0.6	0.6
ED - SPI	262.8	2.1	0.6	0.6
ED - SCI	263.4	2.1	0.6	0.6
ES - 1	264.2	1.9	0.6	0.6
ES - SPI(t)	265.0	1.9	0.5	0.5
ES - SCI(t)	265.8	1.9	0.5	0.5

2.3.3.3. Cost forecasting

BAC	2 244 090.7€	Real Cost	1 868 795.8€	Under Budget	-16.7%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	1995061.0	139184.9	0.8	-0.8
CPI	1709686.1	193451.8	0.9	0.9
SPI	2885137.1	2575356.7	6.0	-6.0
SPI(t)	2085580.6	279210.7	1.3	-1.3
SCI	2372222.7	1871648.3	3.0	-3.0
SCI(t)	1771703.7	203668.4	0.6	0.6
0.8 CPI + 0.2 SPI	1782364.4	182189.4	0.5	0.5
0.8 CPI + 0.2 SPI(t)	1759821.5	163505.1	0.6	0.6